

1

$$\log\left(\frac{p}{1-p}\right) = z = w^T x + b$$

Thus it follows that $p = \frac{1}{1+e^{-z}}$

So

$$p_1 = 0.6456563062, p_2 = 0.0003353501, p_3 = 0.9975273768, p_4 = 0.0000000021, p_5 = 0^*$$

* Note this is due to rounding as the probability in this case asymptotically approaches 0 and never reaches exact 0

Thus the one we are most confident on is z^5 because its probability is furthest from 0.5 and the one we are least confident on is z^1 as its probability is closes to 0.5. We could also look at it as the z 's distance from 0 where further from 0 is more certain and closer to 0 is less certain.

2

We simply need to change our increments in the opposite direction. That is if we get a false positive then we should decrease the weight instead of increasing it. Similarly if we eget a false negative we should increase our weight rather than decrease it. So it follows that we just do the opposite of our steps for a class 1 regression so:

$$\begin{aligned} \frac{dL}{dw_j} &= - \sum_{i=1}^n (\sigma(z^i) - y^i) x_j^i, w_j \leftarrow w_j - \eta \frac{dL}{dw_j} \\ \frac{dL}{db} &= - \sum_{i=1}^n (\sigma(z^i) - y^i), b \leftarrow b - \eta \frac{dL}{db} \end{aligned}$$

For the full derivation we simply replace the $\sigma(z^i)$ with $1 - \sigma(z^i)$ thus:

$$L = \sum_{i=1}^N y^i \log(1 - \sigma(z^i)) + (1 - y^i) \log(\sigma(z^i))$$

For w_j :

$$\begin{aligned} \frac{\delta L}{\delta w_j} &= \sum_{i=1}^N y^i \frac{1}{1-\sigma(z^i)} \sigma(z^i)(1 - \sigma(z^i)) x_j^i + (1 - y^i) \frac{1}{\sigma(z^i)} \sigma(z^i)(1 - \sigma(z^i)) x_j^i \\ \frac{\delta L}{\delta w_j} &= \sum_{i=1}^N y^i \sigma(z^i) x_j^i + (1 - y^i)(1 - \sigma(z^i)) x_j^i \\ \frac{\delta L}{\delta w_j} &= \sum_{i=1}^N y^i \sigma(z^i) x_j^i + (1 - y^i - \sigma(z^i) + y^i \sigma(z^i)) x_j^i \\ \frac{\delta L}{\delta w_j} &= \sum_{i=1}^N (1 - y^i - \sigma(z^i) + 2y^i \sigma(z^i)) x_j^i \end{aligned}$$

For b :

$$\frac{\delta L}{\delta b} = \sum_{i=1}^N y^i \frac{1}{1-\sigma(z^i)} \sigma(z^i)(1-\sigma(z^i)) + (1-y^i) \frac{1}{\sigma(z^i)} \sigma(z^i)(1-\sigma(z^i))$$

$$\frac{\delta L}{\delta b} = \sum_{i=1}^N y^i \sigma(z^i) + (1-y^i)(1-\sigma(z^i))$$

$$\frac{\delta L}{\delta b} = \sum_{i=1}^N y^i \sigma(z^i) + (1-y^i - \sigma(z^i) + y^i \sigma(z^i))$$

$$\frac{\delta L}{\delta b} = \sum_{i=1}^N (1-y^i - \sigma(z^i) + 2y^i \sigma(z^i))$$

Thus it follows that

$$w_j \leftarrow w_j - \eta \frac{dL}{dw_j}$$

$$b \leftarrow b - \eta \frac{dL}{db}$$

3

$$L = \frac{\lambda}{2n} \sum_{i=1}^m w_i^2 + \sum_{i=1}^N y^i \log(\sigma(z^i)) + (1-y^i) \log(1-\sigma(z^i))$$

For w_j :

$$\frac{\delta L}{\delta w_j} = \frac{\lambda}{2n} + \sum_{i=1}^N y^i \frac{1}{1-\sigma(z^i)} \sigma(z^i)(1-\sigma(z^i)) x_j^i + (1-y^i) \frac{1}{\sigma(z^i)} \sigma(z^i)(1-\sigma(z^i)) x_j^i$$

$$\frac{\delta L}{\delta w_j} = \sum_{i=1}^N y^i \sigma(z^i) x_j^i + (1-y^i)(1-\sigma(z^i)) x_j^i + \frac{\lambda}{2n}$$

$$\frac{\delta L}{\delta w_j} = \sum_{i=1}^N y^i \sigma(z^i) x_j^i + (1-y^i - \sigma(z^i) + y^i \sigma(z^i)) x_j^i + \frac{\lambda}{2n}$$

$$\frac{\delta L}{\delta w_j} = \sum_{i=1}^N (1-y^i - \sigma(z^i) + 2y^i \sigma(z^i)) x_j^i + \frac{\lambda}{2n}$$

For b :

$$\frac{\delta L}{\delta b} = \sum_{i=1}^N y^i \frac{1}{1-\sigma(z^i)} \sigma(z^i)(1-\sigma(z^i)) + (1-y^i) \frac{1}{\sigma(z^i)} \sigma(z^i)(1-\sigma(z^i))$$

$$\frac{\delta L}{\delta b} = \sum_{i=1}^N y^i \sigma(z^i) + (1-y^i)(1-\sigma(z^i))$$

$$\frac{\delta L}{\delta b} = \sum_{i=1}^N y^i \sigma(z^i) + (1-y^i - \sigma(z^i) + y^i \sigma(z^i))$$

$$\frac{\delta L}{\delta b} = \sum_{i=1}^N (1-y^i - \sigma(z^i) + 2y^i \sigma(z^i))$$

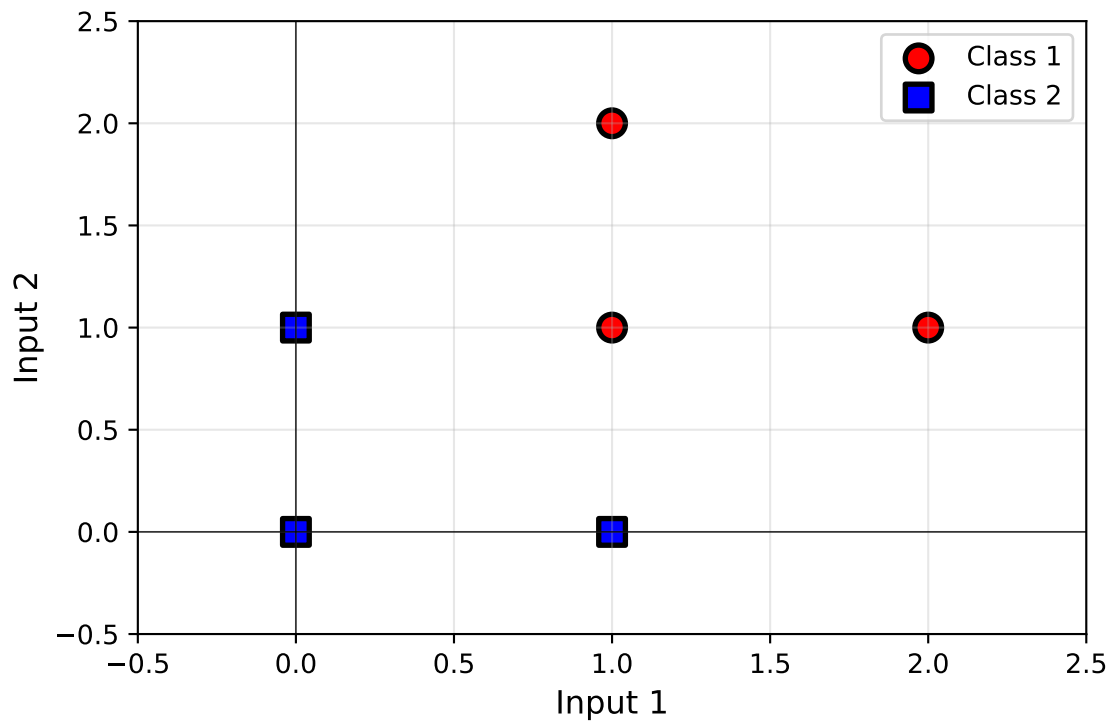
Again it still applies that

$$w_j \leftarrow w_j - \eta \frac{dL}{dw_j}$$

$$b \leftarrow b - \eta \frac{dL}{db}$$

4

a



The dataset is linearly separable by the line $y = 1.5 - x$

b

Decision boundary: $y + x - 1.5 = 0$

Hyperplane: $x + y - 1.5 = \pm 0.5$

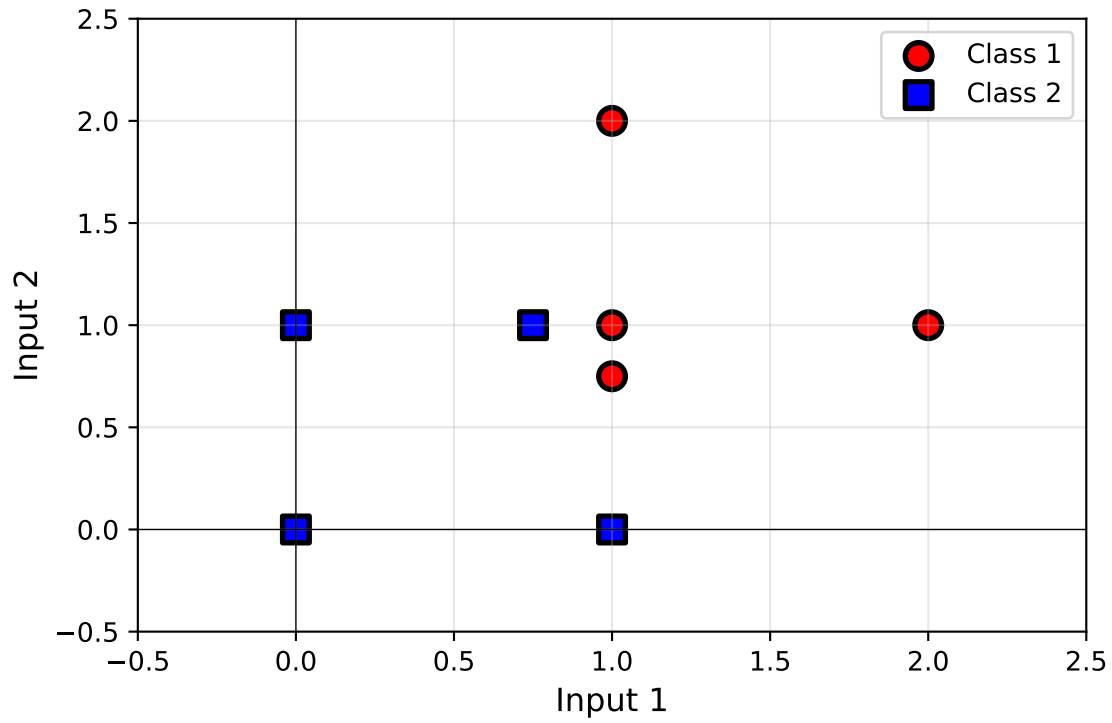
So class 1 hyperplane: $x + y - 2 = 0$

So class 2 hyperplane: $x + y - 1 = 0$

c

The margin is of size 1

5

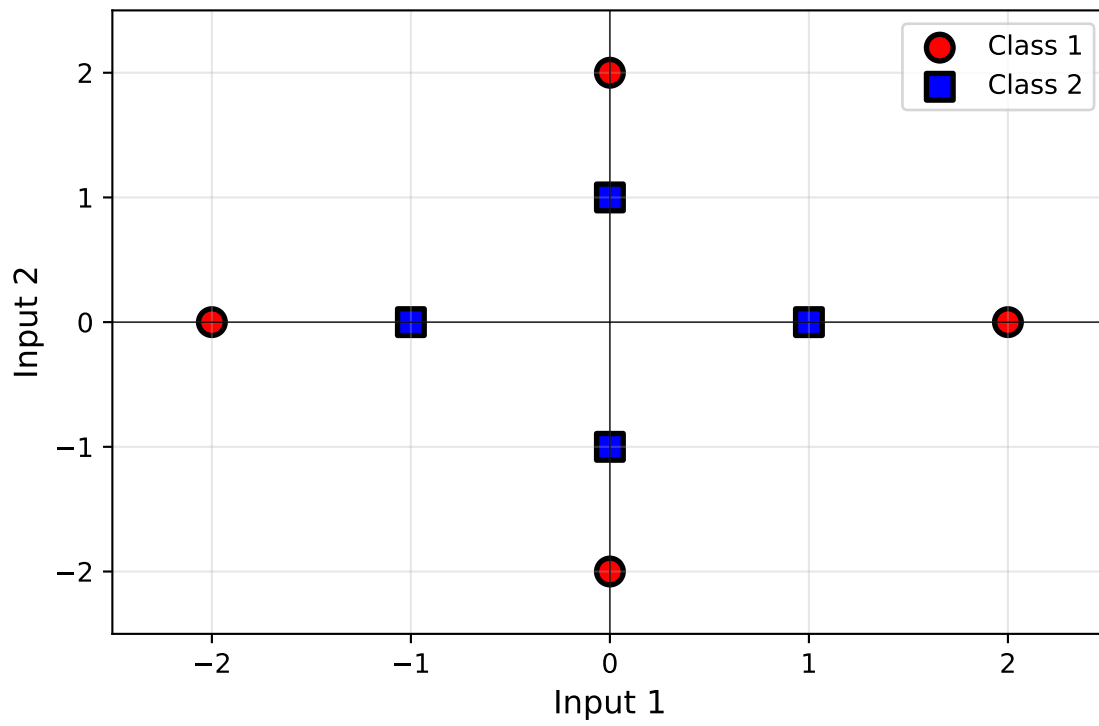


$$\xi_7 = 0.25$$

$$\xi_8 = -0.5$$

6

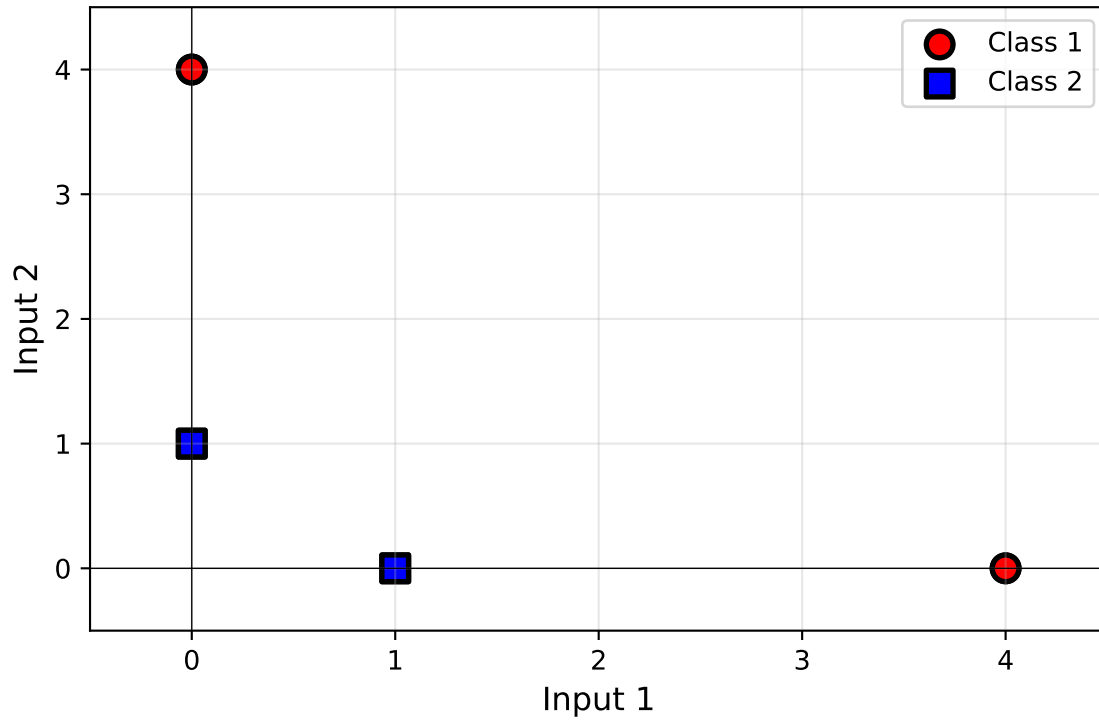
a



This is not linearly separable

b

$$\phi(x, y) = (x^2, y^2)$$



c

7